



Yahui Li

Junior Undergraduate Student

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Education

Tsinghua University

B.S. in Theoretical and Applied Mechanics

Beijing, China

Sep. 2023 - Present

- **Tsien Excellence in Engineering Program**
- **GPA: 3.90/4.00**
- **Core Academic Training:** Fluid Mechanics, Viscous Fluid Dynamics, Solid Mechanics, Convex Optimization, Mathematical Principles of Neuroscience.

Research Experience

Research Intern, Event-based Daily Human Action Recognition with On-chip Learning

Tsinghua University

2024/09 – 2025/09

Beijing, China

Advisor: Prof. Hong Chen

- Constructed a low-latency event-data pipeline, performing pre-processing denoising and neuromorphic encoding for robust action perception under noisy sensory input.
- Implemented hardware–software co-design and FPGA-based data transmission, designing data protocols, FIFO buffer interaction to support real-time system demonstration.
- **First-author demo paper under review (ISCAS 2026) demonstrating the system’s adaptive on-chip learning, which achieves higher-accuracy action recognition by adapting to different camera angles.**

ISURE Program

University of Notre Dame

2025/07 – 2025/12

Notre Dame, IN, USA

Advisor: Prof. Jianxun Wang

- Investigated neural field models for turbulence data dimension reduction; **performed ablation studies for the manuscript *Conditional Neural Field for Spatial Dimension Reduction of Turbulence Data: A Comparison Study***, under review at *Physics of Fluids*.
- Explored reinforcement-learning-based control strategies for fish-inspired locomotion, analyzing vortex-mediated sensing and unsteady flow signatures and Kolmogorov-Sivashinsky equation control.
- Conducted PDE closure modeling to understand nonlinear dynamics and information structures in turbulent flows.

Undergraduate Researcher, Insect Olfaction and Flapping-Wing Active Sampling

Tsinghua University

2025/09 – Present

Beijing, China

Advisor: Prof. Weixi Huang

- Initiated the project ***Insect Olfaction and Flapping-Wing Active Sampling: Spatiotemporal Sensing Mechanism of Turbulent Plumes***.
- Modeled insect wing geometry and flapping kinematics to construct a biologically real actuation framework.
- Developed a three-dimensional scalar-transport solver integrated into an existing velocity solver for plume–flow simulations.
- Investigating vortex–odor interactions and identifying flow structures that encode chemical cues for bio-inspired active sensing strategies.

Awards

2023	Excellent Presentation Award , 13th Creative Competition & 2023 Tsinghua Student Creative Challenge	Tsinghua University
2024	Comprehensive Excellence Scholarship , Department-level Merit-based Scholarship	Tsinghua University
2024	Tsinghua Technology Taihu Scholarship , For outstanding students in science and technology innovation	Tsinghua University
2025	Tsinghua Technology Taihu Scholarship , For outstanding students in science and technology innovation	Tsinghua University

Skills

- **Programming:** Python, Fortran, Jax, PyTorch
- **Languages:** Mandarin (native), English (fluent)
- **Interests:** Latin dance, modern dance (ballroom)